

7 (a) Some of the energy levels of a particular atom X are shown in Fig. 7.1.

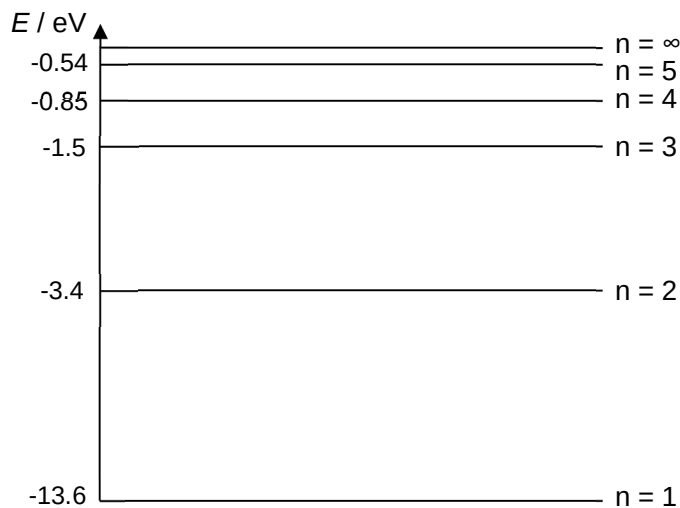


Fig. 7.1

(i) State the ionisation energy of atom X.

ionisation energy = eV [1]

(ii) Cool vapour of X at low pressure is bombarded with electrons of kinetic energy 2.00×10^{-18} J.

Determine the number of different wavelengths of radiation that will be emitted.

number of wavelengths = [2]

(iii) The electrons in (ii) were replaced with photons of the same energy to bombard the cool vapour of X.

Suggest the number of different wavelengths of radiation that will be emitted. Explain your answer.

.....

 [2]

- (b) An X-ray spectrum is first produced by an X-ray tube with tungsten. Another X-ray spectrum is then produced using barium and both spectra are shown in Fig. 7.2.

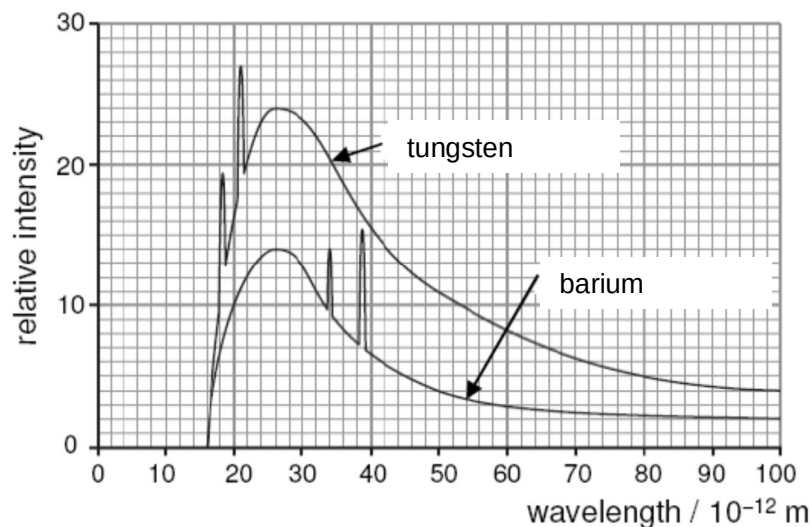


Fig. 7.2

- (i) The accelerating potential used to produce the X-ray spectra using tungsten and barium are the same.

By reference to Fig. 7.2, state a feature how this can be deduced.

..... [1]

- (ii) Hence, determine the accelerating potential.

accelerating potential = V [2]

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Section B

Answer **one** question from this section.

8 (a) Explain why a body moving with uniform speed in a circle must experience a force